• 卡尔曼滤波公式与计算步骤

先验估计

$$\hat{x}_{k|k-1} = A\hat{x}_{k-1|k-1} + Bu_k$$

先验误差矩阵

$$P_{k|k-1} = AP_{k-1|k-1}A^T + Q$$

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$$K_k = P_{k|k-1}H^T(HP_{k|k-1}H^T + R)^{-1}$$

后验估计

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(z_k - H\hat{x}_{k|k-1})$$

更新协方差矩阵

$$P_{k|k} = (I - K_kH)P_{k|k-1}$$

其中$A, B, H, Q, R, \hat{x}_{0|0}, P_{0|0}$需自行给定。

• 扩展卡尔曼滤波

扩展卡尔曼滤波在形式上与卡尔曼滤波类似，但是它事实上不能直接从卡尔曼滤波的数学公式推出来，我们这里不去推导理论公式，只说明如何将它运用在无感 FOC 控制中

卡尔曼滤波使用线性方程（下左），而扩展卡尔曼滤波使用非线性方程（下右）：

$$\begin{cases} x_k = Ax_{k-1} + Bu_k + w_k \\ z_k = Hx_k + v_k \end{cases}, \begin{cases} x_k = f(x_{k-1}, u_k) + w_k \\ z_k = h(x_k) + v_k \end{cases}$$

由于我们的u_k项和$h(x_k)$项为线性的，我们也把上述式子写成：

$$\begin{cases} x_k = f(x_{k-1}) + Bu_k \\ y_k = Cx_k \end{cases}$$

其中w_k和v_k为过程噪声和测量噪声，仅在卡尔曼滤波推导过程中使用，我们只是使用该公式，不需要带上这两项。另外，输出变量在理论推导时我们写成z_k，使用时一般写成y_k

永磁同步电机的静止坐标系中的定子电压方程为：

$$\begin{cases} u_\alpha = Ri_\alpha + \dfrac{d\psi_\alpha}{dt} = Ri_\alpha + L_s\dfrac{di_\alpha}{dt} - \omega_e\psi_f sin\theta_e \\ u_\beta = Ri_\beta + \dfrac{d\psi_\beta}{dt} = Ri_\beta + L_s\dfrac{di_\beta}{dt} + \omega_e\psi_f cos\theta_e \end{cases}$$

上述方程可以变换成：

$$\begin{cases} \dfrac{di_\alpha}{dt} = \dfrac{u_\alpha}{L_s} - \dfrac{R}{L_s}i_\alpha + \dfrac{1}{L_s}\omega_e\psi_f sin\theta_e \\ \dfrac{di_\beta}{dt} = \dfrac{u_\beta}{L_s} - \dfrac{R}{L_s}i_\beta - \dfrac{1}{L_s}\omega_e\psi_f cos\theta_e \\ \dfrac{d\omega_e}{dt} = 0 \\ \dfrac{d\theta_e}{dt} = \omega_e \end{cases}$$

其中$\frac{d\omega_e}{dt} = 0$ 和$\frac{d\theta_e}{dt} = \omega_e$是我们自己加上去的，我们使用了简化的模型，认为$\frac{d\omega_e}{dt} = 0$ 即ω_e变化缓慢。

输入变量，状态变量，输出变量我们取：

$$u=\begin{bmatrix}u_\alpha\\u_\beta\end{bmatrix},x=\begin{bmatrix}i_\alpha\\i_\beta\\\omega_e\\\theta_e\end{bmatrix},y=\begin{bmatrix}i_\alpha\\i_\beta\end{bmatrix}$$

得到：

$$\dot{x}_k=g(x_{k-1},u_k),其中g(x)=\begin{bmatrix}-\frac{R}{L_s}i_\alpha+\frac{1}{L_s}\omega_e\psi_f sin\theta_e+\frac{1}{L_s}u_0\\-\frac{R}{L_s}i_\beta-\frac{1}{L_s}\omega_e\psi_f cos\theta_e+\frac{1}{L_s}u_1\\0\\\omega_e\end{bmatrix}$$

使用前向欧拉离散化（EKF 执行周期设为T_s）：

$$\frac{x_k-x_{k-1}}{T_s}\approx\dot{x}_k=g(x_{k-1},u_k)$$

$$x_k=x_{k-1}+T_s g(x_{k-1},u_k)$$

把上述离散方程写成以下形式：

$$x_k=f(x_{k-1},u_k),y_k=Cx_k$$

其中：

$$f(x,u)=x+T_s g(x,u),H=C=\begin{bmatrix}1&0&0&0\\0&1&0&0\end{bmatrix}$$

我们本来拥有$\hat{x}_{k-1|k-1}$，可以直接算出先验估计

$$\hat{x}_{k|k-1}=f(\hat{x}_{k-1|k-1},u_k)$$

但是在计算先验误差矩阵时

$$P_{k|k-1}=AP_{k-1|k-1}A^T+Q$$

我们缺少矩阵 A，扩展卡尔曼滤波的做法是使用雅克比矩阵代替 A：

$$A=\frac{\partial f}{\partial x}\Big|_{(\hat{x}_{k-1|k-1},u_k)}=\begin{bmatrix}1-T_s\frac{R}{L_s}&0&T_s\frac{1}{L_s}\psi_f sin\theta_e&T_s\frac{1}{L_s}\omega_e\psi_f cos\theta_e\\0&1-T_s\frac{R}{L_s}&-T_s\frac{1}{L_s}\psi_f cos\theta_e&T_s\frac{1}{L_s}\omega_e\psi_f sin\theta_e\\0&0&1&0\\0&0&T_s&1\end{bmatrix}_{\Big|(\hat{x}_{k-1|k-1},u_k)}$$

$$=\begin{bmatrix}1-T_s\frac{R}{L_s}&0&T_s\frac{1}{L_s}\psi_f sin\theta_{e(k-1)}&T_s\frac{1}{L_s}\omega_{e(k-1)}\psi_f cos\theta_{e(k-1)}\\0&1-T_s\frac{R}{L_s}&-T_s\frac{1}{L_s}\psi_f cos\theta_{e(k-1)}&T_s\frac{1}{L_s}\omega_{e(k-1)}\psi_f sin\theta_{e(k-1)}\\0&0&1&0\\0&0&T_s&1\end{bmatrix}$$

此时可计算先验误差矩阵

$$P_{k|k-1}=AP_{k-1|k-1}A^T+Q$$

剩下的直接套公式即可：

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$$K_k=P_{k|k-1}H^T(HP_{k|k-1}H^T+R)^{-1}=P_{k|k-1}C^T(CP_{k|k-1}C^T+R)^{-1}$$

后验估计

$$\hat{x}_{k|k}=\hat{x}_{k|k-1}+K_k(z_k-h(\tilde{x}_{k-1},0))=\hat{x}_{k|k-1}+K_k(y_k-C\tilde{x}_{k-1})$$

其中

$$\tilde{x}_{k-1} = f(\hat{x}_{k-1|k-1}, u_k) = \hat{x}_{k|k-1}$$

更新协方差矩阵

$$P_{k|k} = (I - K_k C)P_{k|k-1}$$

其中

$$y_k = z_k, B = T_s\begin{bmatrix} \frac{1}{L_s} & 0 \\ 0 & \frac{1}{L_s} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, H = C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

• 矩阵展开和优化

由于直接使用 for for for 来计算矩阵非常的耗时，一次运算需要接近 200us，满足不了 20khz 的 FOC（50us 周期），因此需要手动展开矩阵

遵循两个原则：实对称矩阵只存一半的变量，矩阵中为 0 的变量不存储，为常数的变量选择性地存储，常数的乘除法可以合并的就合并，提前计算好，接下来将一个个推导和编写代码

先验估计

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1|k-1}, u_k)$$

$$\hat{x}_{k|k-1} = \begin{bmatrix} i_{\alpha(k-1)} \\ i_{\beta(k-1)} \\ \omega_{e(k-1)} \\ \theta_{e(k-1)} \end{bmatrix} + T_s\begin{bmatrix} -\frac{R}{L_s}i_{\alpha(k-1)} + \frac{1}{L_s}\omega_{e(k-1)}\psi_f sin\theta_{e(k-1)} + \frac{1}{L_s}u_{k-1}[0] \\ -\frac{R}{L_s}i_{\beta(k-1)} - \frac{1}{L_s}\omega_{e(k-1)}\psi_f cos\theta_{e(k-1)} + \frac{1}{L_s}u_{k-1}[1] \\ 0 \\ \omega_{e(k-1)} \end{bmatrix}$$

```
const float i_alpha_k_prev = x_k_k[0];
const float i_beta_k_prev = x_k_k[1];
const float w_e_k_prev = x_k_k[2];
const float theta_e_k_prev = x_k_k[3];

float c_theta_e_k_prev,s_theta_e_k_prev;//不能为const

static const float RAD_TO_DEG = 180.0f / PI;
arm_sin_cos_f32(theta_e_k_prev * RAD_TO_DEG, &s_theta_e_k_prev,
&c_theta_e_k_prev);
const float w_flux = w_e_k_prev * flux;
x_k_k_prev[0] = i_alpha_k_prev +
                    (-Rs    * i_alpha_k_prev + w_flux *s_theta_e_k_prev+
ekf->Valpha_I )*TsDivLs;
x_k_k_prev[1] = i_beta_k_prev +
                    (-Rs * i_beta_k_prev - w_flux * c_theta_e_k_prev +
ekf->Vbeta_I)*TsDivLs;
x_k_k_prev[2] = w_e_k_prev;
x_k_k_prev[3] = theta_e_k_prev + T_s * w_e_k_prev;
```

Ak 矩阵

$$A_k = \frac{\partial f}{\partial x}_{|(\hat{x}_{k-1|k-1},u_k)}$$

$$= \begin{bmatrix} 1 - T_s\frac{R}{L_s} & 0 & T_s\frac{1}{L_s}\psi_f sin\theta_{e(k-1)} & T_s\frac{1}{L_s}\omega_{e(k-1)}\psi_f cos\theta_{e(k-1)} \\ 0 & 1 - T_s\frac{R}{L_s} & -T_s\frac{1}{L_s}\psi_f cos\theta_{e(k-1)} & T_s\frac{1}{L_s}\omega_{e(k-1)}\psi_f sin\theta_{e(k-1)} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & T_s & 1 \end{bmatrix}$$

```
const float A_k_0_0=1 - Rs*TsDivLs;
const float A_k_1_1=1 - Rs*TsDivLs;
const float A_k_3_2=T_s;
const float TsDivLsFlux=TsDivLs*flux;
const float A_k_0_2= TsDivLsFlux*s_theta_e_k_prev;
const float A_k_1_2= - TsDivLsFlux * c_theta_e_k_prev;
const float A_k_0_3= -A_k_1_2*w_e_k_prev;
const float   A_k_1_3 = A_k_0_2*w_e_k_prev;
```

先验方差矩阵

$$P_{k|k-1} = A_k P_{k-1|k-1} A_k^T + W_k Q W_k^T = A_k P_{k-1|k-1} A_k^T + Q$$

这里设 Q 为对角矩阵，然后设定$P_{0|0}$为实对称阵，根据上面的公式可以递推得出$P_{k|k-1}$也为实对称阵。

$$P_{k|k-1} = \begin{bmatrix} a_{k(0,0)} & 0 & a_{k(0,2)} & a_{k(0,3)} \\ 0 & a_{k(1,1)} & a_{k(1,2)} & a_{k(1,3)} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & a_{k(3,2)} & 1 \end{bmatrix}$$

$$* \begin{bmatrix} P_{k-1|k-1(0,0)} & P_{k-1|k-1(0,1)} & P_{k-1|k-1(0,2)} & P_{k-1|k-1(0,3)} \\ P_{k-1|k-1(0,1)} & P_{k-1|k-1(1,1)} & P_{k-1|k-1(1,2)} & P_{k-1|k-1(1,3)} \\ P_{k-1|k-1(0,2)} & P_{k-1|k-1(1,2)} & P_{k-1|k-1(2,2)} & P_{k-1|k-1(2,3)} \\ P_{k-1|k-1(0,3)} & P_{k-1|k-1(1,3)} & P_{k-1|k-1(2,3)} & P_{k-1|k-1(3,3)} \end{bmatrix} A_k^T + Q$$

使用分块矩阵计算$A_k P_{k-1|k-1} A_k^T$：

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}\begin{bmatrix} P_A & P_B \\ P_C & P_D \end{bmatrix}\begin{bmatrix} A^T & C^T \\ B^T & D^T \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & D \end{bmatrix}\begin{bmatrix} P_A & P_B \\ P_C & P_D \end{bmatrix}\begin{bmatrix} A & 0 \\ B^T & D^T \end{bmatrix}$$

$$= \begin{bmatrix} AP_A + BP_C & AP_B + BP_D \\ DP_C & DP_D \end{bmatrix}\begin{bmatrix} A & 0 \\ B^T & D^T \end{bmatrix}$$

$$= \begin{bmatrix} AP_AA + BP_CA + AP_BB^T + BP_DB^T & AP_BD^T + BP_DD^T \\ DP_CA + DP_DB^T & DP_DD^T \end{bmatrix}$$

$$= \begin{bmatrix} AP_AA + BP_B^TA + AP_BB^T + BP_DB^T & AP_BD^T + BP_DD^T \\ DP_CA + DP_DB^T & DP_DD^T \end{bmatrix}$$

由于上述计算结果仍为对称矩阵，因此我们只需要计算右上角的三个值（其实即$(DP_CA + DP_DB^T)^T = AP_BD^T + BP_DD^T$，可以少算一个）。

$$AP_AA + BP_B^TA + AP_BB^T + BP_DB^T = AP_AA + BP_B^TA + (BP_B^TA)^T + BP_DB^T$$

$$AP_AA=\begin{bmatrix} a_{k(0,0)} & 0 \\ 0 & a_{k(1,1)} \end{bmatrix}\begin{bmatrix} P_{k-1|k-1(0,0)} & P_{k-1|k-1(0,1)} \\ P_{k-1|k-1(0,1)} & P_{k-1|k-1(1,1)} \end{bmatrix}\begin{bmatrix} a_{k(0,0)} & 0 \\ 0 & a_{k(1,1)} \end{bmatrix}$$

$$=\begin{bmatrix} a_{k(0,0)}P_{k-1|k-1(0,0)} & a_{k(0,0)}P_{k-1|k-1(0,1)} \\ a_{k(1,1)}P_{k-1|k-1(0,1)} & a_{k(1,1)}P_{k-1|k-1(1,1)} \end{bmatrix}\begin{bmatrix} a_{k(0,0)} & 0 \\ 0 & a_{k(1,1)} \end{bmatrix}$$

$$=\begin{bmatrix} a_{k(0,0)}a_{k(0,0)}P_{k-1|k-1(0,0)} & a_{k(0,0)}a_{k(1,1)}P_{k-1|k-1(0,1)} \\ a_{k(0,0)}a_{k(1,1)}P_{k-1|k-1(0,1)} & a_{k(1,1)}a_{k(1,1)}P_{k-1|k-1(1,1)} \end{bmatrix}$$

$$BP_DB^T=\begin{bmatrix} a_{k(0,2)} & a_{k(0,3)} \\ a_{k(1,2)} & a_{k(1,3)} \end{bmatrix}\begin{bmatrix} P_{k-1|k-1(2,2)} & P_{k-1|k-1(2,3)} \\ P_{k-1|k-1(2,3)} & P_{k-1|k-1(3,3)} \end{bmatrix}\begin{bmatrix} a_{k(0,2)} & a_{k(1,2)} \\ a_{k(0,3)} & a_{k(1,3)} \end{bmatrix}$$

$$=\begin{bmatrix} a_{k(0,2)}P_{k-1|k-1(2,2)}+a_{k(0,3)}P_{k-1|k-1(2,3)} & a_{k(0,2)}P_{k-1|k-1(2,3)}+a_{k(0,3)}P_{k-1|k-1(3,3)} \\ a_{k(1,2)}P_{k-1|k-1(2,2)}+a_{k(1,3)}P_{k-1|k-1(2,3)} & a_{k(1,2)}P_{k-1|k-1(2,3)}+a_{k(1,3)}P_{k-1|k-1(3,3)} \end{bmatrix}\begin{bmatrix} a_{k(0,2)} & a_{k(1,2)} \\ a_{k(0,3)} & a_{k(1,3)} \end{bmatrix}$$

$$=\begin{bmatrix} BPB_{(0,0)} & BPB_{(0,1)} \\ BPB_{(1,0)} & BPB_{(1,1)} \end{bmatrix}$$

$$BPB_{(0,0)}=a_{k(0,2)}a_{k(0,2)}P_{k-1|k-1(2,2)}+2a_{k(0,2)}a_{k(0,3)}P_{k-1|k-1(2,3)}+a_{k(0,3)}a_{k(0,3)}P_{k-1|k-1(3,3)}$$

$$\begin{aligned} BPB_{(0,1)}=BPB_{(1,0)} \\ &=a_{k(1,2)}(a_{k(0,2)}P_{k-1|k-1(2,2)}+a_{k(0,3)}P_{k-1|k-1(2,3)})+a_{k(1,3)}(a_{k(0,2)}P_{k-1|k-1(2,3)} \\ &+a_{k(0,3)}P_{k-1|k-1(3,3)}) \end{aligned}$$

$$BPB_{(1,1)}=a_{k(1,2)}a_{k(1,2)}P_{k-1|k-1(2,2)}+2a_{k(1,2)}a_{k(1,3)}P_{k-1|k-1(2,3)}+a_{k(1,3)}a_{k(1,3)}P_{k-1|k-1(3,3)}$$

$$BP_B^TA=\begin{bmatrix} a_{k(0,2)} & a_{k(0,3)} \\ a_{k(1,2)} & a_{k(1,3)} \end{bmatrix}\begin{bmatrix} P_{k-1|k-1(0,2)} & P_{k-1|k-1(1,2)} \\ P_{k-1|k-1(0,3)} & P_{k-1|k-1(1,3)} \end{bmatrix}\begin{bmatrix} a_{k(0,0)} & 0 \\ 0 & a_{k(1,1)} \end{bmatrix}$$

$$=\begin{bmatrix} a_{k(0,2)}P_{k-1|k-1(0,2)}+a_{k(0,3)}P_{k-1|k-1(0,3)} & a_{k(0,2)}P_{k-1|k-1(1,2)}+a_{k(0,3)}P_{k-1|k-1(1,3)} \\ a_{k(1,2)}P_{k-1|k-1(0,2)}+a_{k(1,3)}P_{k-1|k-1(0,3)} & a_{k(1,2)}P_{k-1|k-1(1,2)}+a_{k(1,3)}P_{k-1|k-1(1,3)} \end{bmatrix}\begin{bmatrix} a_{k(0,0)} & 0 \\ 0 & a_{k(1,1)} \end{bmatrix}$$

$$=\begin{bmatrix} a_{k(0,0)}(a_{k(0,2)}P_{k-1|k-1(0,2)}+a_{k(0,3)}P_{k-1|k-1(0,3)}) & a_{k(1,1)}(a_{k(0,2)}P_{k-1|k-1(1,2)}+a_{k(0,3)}P_{k-1|k-1(1,3)}) \\ a_{k(0,0)}(a_{k(1,2)}P_{k-1|k-1(0,2)}+a_{k(1,3)}P_{k-1|k-1(0,3)}) & a_{k(1,1)}(a_{k(1,2)}P_{k-1|k-1(1,2)}+a_{k(1,3)}P_{k-1|k-1(1,3)}) \end{bmatrix}$$

因此结果为：

$$AP_AA+BP_B^TA+AP_BB^T+BP_DB^T=\begin{bmatrix} P_{k|k-1(0,0)} & P_{k-1|k-1(0,1)} \\ P_{k|k-1(0,1)} & P_{k|k-1(1,1)} \end{bmatrix}$$

$$\begin{aligned} P_{k|k-1(0,0)} &=a_{k(0,0)}a_{k(0,0)}P_{k-1|k-1(0,0)}+2a_{k(0,0)}(a_{k(0,2)}P_{k-1|k-1(0,2)}+a_{k(0,3)}P_{k-1|k-1(0,3)}) \\ &+a_{k(0,2)}a_{k(0,2)}P_{k-1|k-1(2,2)}+2a_{k(0,2)}a_{k(0,3)}P_{k-1|k-1(2,3)} \\ &+a_{k(0,3)}a_{k(0,3)}P_{k-1|k-1(3,3)} \end{aligned}$$

$$\begin{aligned} P_{k|k-1(0,1)} &=a_{k(0,0)}a_{k(1,1)}P_{k-1|k-1(0,1)}+a_{k(1,1)}(a_{k(0,2)}P_{k-1|k-1(1,2)}+a_{k(0,3)}P_{k-1|k-1(1,3)}) \\ &+a_{k(0,0)}(a_{k(1,2)}P_{k-1|k-1(0,2)}+a_{k(1,3)}P_{k-1|k-1(0,3)})+a_{k(1,2)}(a_{k(0,2)}P_{k-1|k-1(2,2)} \\ &+a_{k(0,3)}P_{k-1|k-1(2,3)})+a_{k(1,3)}(a_{k(0,2)}P_{k-1|k-1(2,3)}+a_{k(0,3)}P_{k-1|k-1(3,3)}) \end{aligned}$$

$$\begin{aligned} P_{k|k-1(1,1)} &=a_{k(1,1)}a_{k(1,1)}P_{k-1|k-1(1,1)}+2a_{k(1,1)}(a_{k(1,2)}P_{k-1|k-1(1,2)}+a_{k(1,3)}P_{k-1|k-1(1,3)}) \\ &+a_{k(1,2)}a_{k(1,2)}P_{k-1|k-1(2,2)}+2a_{k(1,2)}a_{k(1,3)}P_{k-1|k-1(2,3)} \\ &+a_{k(1,3)}a_{k(1,3)}P_{k-1|k-1(3,3)} \end{aligned}$$

$$(AP_B+BP_D)D^T$$

$$AP_B + BP_D$$
$$= \begin{bmatrix} a_{k(0,0)} & 0 \\ 0 & a_{k(1,1)} \end{bmatrix} \begin{bmatrix} P_{k-1|k-1(0,2)} & P_{k-1|k-1(0,3)} \\ P_{k-1|k-1(1,2)} & P_{k-1|k-1(1,3)} \end{bmatrix}$$
$$+ \begin{bmatrix} a_{k(0,2)} & a_{k(0,3)} \\ a_{k(1,2)} & a_{k(1,3)} \end{bmatrix} \begin{bmatrix} P_{k-1|k-1(2,2)} & P_{k-1|k-1(2,3)} \\ P_{k-1|k-1(2,3)} & P_{k-1|k-1(3,3)} \end{bmatrix}$$
$$= \begin{bmatrix} a_{k(0,0)}P_{k-1|k-1(0,2)} & a_{k(0,0)}P_{k-1|k-1(0,3)} \\ a_{k(1,1)}P_{k-1|k-1(1,2)} & a_{k(1,1)}P_{k-1|k-1(1,3)} \end{bmatrix}$$
$$+ \begin{bmatrix} a_{k(0,2)}P_{k-1|k-1(2,2)} + a_{k(0,3)}P_{k-1|k-1(2,3)} & a_{k(0,2)}P_{k-1|k-1(2,3)} + a_{k(0,3)}P_{k-1|k-1(3,3)} \\ a_{k(1,2)}P_{k-1|k-1(2,2)} + a_{k(1,3)}P_{k-1|k-1(2,3)} & a_{k(1,2)}P_{k-1|k-1(2,3)} + a_{k(1,3)}P_{k-1|k-1(3,3)} \end{bmatrix}$$
$$(AP_B + BP_D)D^T = \begin{bmatrix} APpBP_{(0,0)} & APpBP_{(0,1)} \\ APpBP_{(1,0)} & APpBP_{(1,1)} \end{bmatrix} \begin{bmatrix} 1 & a_{k(3,2)} \\ 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} APpBP_{(0,0)} & a_{k(3,2)}APpBP_{(0,0)} + APpBP_{(0,1)} \\ APpBP_{(1,0)} & a_{k(3,2)}APpBP_{(1,0)} + APpBP_{(1,1)} \end{bmatrix}$$
$$DP_DD^T = \begin{bmatrix} 1 & 0 \\ a_{k(3,2)} & 1 \end{bmatrix} \begin{bmatrix} P_{k-1|k-1(2,2)} & P_{k-1|k-1(2,3)} \\ P_{k-1|k-1(2,3)} & P_{k-1|k-1(3,3)} \end{bmatrix} \begin{bmatrix} 1 & a_{k(3,2)} \\ 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} P_{k-1|k-1(2,2)} & P_{k-1|k-1(2,3)} \\ a_{k(3,2)}P_{k-1|k-1(2,2)} + P_{k-1|k-1(2,3)} & a_{k(3,2)}P_{k-1|k-1(2,3)} + P_{k-1|k-1(3,3)} \end{bmatrix} \begin{bmatrix} 1 & a_{k(3,2)} \\ 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} P_{k-1|k-1(2,2)} & a_{k(3,2)}P_{k-1|k-1(2,2)} + P_{k-1|k-1(2,3)} \\ a_{k(3,2)}P_{k-1|k-1(2,2)} + P_{k-1|k-1(2,3)} & a_{k(3,2)}a_{k(3,2)}P_{k-1|k-1(2,2)} + 2a_{k(3,2)}P_{k-1|k-1(2,3)} + P_{k-1|k-1(3,3)} \end{bmatrix}$$

经过推导，我们可以写出代码如下

```
//计算P_k_k_prev

P_k_k_prev_0_0=A_k_0_0*A_k_0_0*P_k_k_0_0+2*A_k_0_0*(A_k_0_2*P_k_k_0_2+A_k_0_3*P_k_k_0_3)

+A_k_0_2*A_k_0_2*P_k_k_2_2+2*A_k_0_2*A_k_0_3*P_k_k_2_3+A_k_0_3*A_k_0_3*P_k_k_3_3+Q_0_0;
P_k_k_prev_0_1=A_k_0_0*A_k_1_1*P_k_k_0_1+A_k_1_1*(A_k_0_2*P_k_k_1_2+A_k_0_3*P_k_k_1_3)
                    +A_k_0_0*(A_k_1_2*P_k_k_0_2+A_k_1_3*P_k_k_0_3)
                    +A_k_1_2*(A_k_0_2*P_k_k_2_2+A_k_0_3*P_k_k_2_3)
                    +A_k_1_3*(A_k_0_2*P_k_k_2_3+A_k_0_3*P_k_k_3_3);
P_k_k_prev_1_1=A_k_1_1*A_k_1_1*P_k_k_1_1+2*A_k_1_1*(A_k_1_2*P_k_k_1_2+A_k_1_3*P_k_k_1_3)
                    +A_k_1_2*A_k_1_2*P_k_k_2_2+2*A_k_1_2*A_k_1_3*P_k_k_2_3
                    +A_k_1_3*A_k_1_3*P_k_k_3_3+Q_1_1;
P_k_k_prev_2_2=P_k_k_2_2+Q_2_2;
P_k_k_prev_3_3=A_k_3_2*A_k_3_2*P_k_k_2_2+2*A_k_3_2*P_k_k_2_3+P_k_k_3_3+Q_3_3;
P_k_k_prev_2_3=A_k_3_2*P_k_k_2_2+P_k_k_2_3;

const float
APpBP_0_0=A_k_0_0*P_k_k_0_2+A_k_0_2*P_k_k_2_2+A_k_0_3*P_k_k_2_3
,APpBP_0_1=A_k_0_0*P_k_k_0_3+A_k_0_2*P_k_k_2_3+A_k_0_3*P_k_k_3_3
```

,APpBP_1_0=A_k_1_1*P_k_k_1_2+A_k_1_2*P_k_k_2_2+A_k_1_3*P_k_k_2_3
,APpBP_1_1=A_k_1_1*P_k_k_1_3+A_k_1_2*P_k_k_2_3+A_k_1_3*P_k_k_3_3;
P_k_k_prev_0_2=APpBP_0_0;
P_k_k_prev_1_2=APpBP_1_0;
P_k_k_prev_0_3=A_k_3_2*APpBP_0_0+APpBP_0_1;
P_k_k_prev_1_3=A_k_3_2*APpBP_1_0+APpBP_1_1;

卡尔曼增益

$$K_k = P_{k|k-1}H^T(HP_{k|k-1}H^T + R)^{-1} = P_{k|k-1}C^T(CP_{k|k-1}C^T + R)^{-1}, C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$P_{k|k-1}C^T = \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} & P_{k|k-1(0,2)} & P_{k|k-1(0,3)} \\ P_{k|k-1(0,1)} & P_{k|k-1(1,1)} & P_{k|k-1(1,2)} & P_{k|k-1(1,3)} \\ P_{k|k-1(0,2)} & P_{k|k-1(1,2)} & P_{k|k-1(2,2)} & P_{k|k-1(2,3)} \\ P_{k|k-1(0,3)} & P_{k|k-1(1,3)} & P_{k|k-1(2,3)} & P_{k|k-1(3,3)} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} \\ P_{k|k-1(0,1)} & P_{k|k-1(1,1)} \\ P_{k|k-1(0,2)} & P_{k|k-1(1,2)} \\ P_{k|k-1(0,3)} & P_{k|k-1(1,3)} \end{bmatrix}$$

$$CP_{k|k-1}C^T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} \\ P_{k|k-1(0,1)} & P_{k|k-1(1,1)} \\ P_{k|k-1(0,2)} & P_{k|k-1(1,2)} \\ P_{k|k-1(0,3)} & P_{k|k-1(1,3)} \end{bmatrix} = \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} \\ P_{k|k-1(0,1)} & P_{k|k-1(1,1)} \end{bmatrix}$$

$$(CP_{k|k-1}C^T + R)^{-1} = \begin{bmatrix} P_{k|k-1(0,0)} + R_{(0,0)} & P_{k|k-1(0,1)} \\ P_{k|k-1(0,1)} & P_{k|k-1(1,1)} + R_{(1,1)} \end{bmatrix}^{-1}$$

行列式（这里理论分析其必定可逆，因此行列式必定不为 0）

$$\left|CP_{k|k-1}C^T + R\right| = (P_{k|k-1(0,0)} + R_{(0,0)})(P_{k|k-1(1,1)} + R_{(1,1)}) - {P_{k|k-1(0,1)}}^2$$

$$(CP_{k|k-1}C^T + R)^{-1} = \frac{1}{\left|CP_{k|k-1}C^T + R\right|} \begin{bmatrix} P_{k|k-1(1,1)} + R_{(1,1)} & -P_{k|k-1(0,1)} \\ -P_{k|k-1(0,1)} & P_{k|k-1(0,0)} + R_{(0,0)} \end{bmatrix}$$

$$K_k = P_{k|k-1}C^T(CP_{k|k-1}C^T + R)^{-1}$$

$$= \frac{1}{\left|CP_{k|k-1}C^T + R\right|} \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} \\ P_{k|k-1(0,1)} & P_{k|k-1(1,1)} \\ P_{k|k-1(0,2)} & P_{k|k-1(1,2)} \\ P_{k|k-1(0,3)} & P_{k|k-1(1,3)} \end{bmatrix} \begin{bmatrix} P_{k|k-1(1,1)} + R_{(1,1)} & -P_{k|k-1(0,1)} \\ -P_{k|k-1(0,1)} & P_{k|k-1(0,0)} + R_{(0,0)} \end{bmatrix}$$

$$= \frac{1}{\left|CP_{k|k-1}C^T + R\right|}$$

$$* \begin{bmatrix} P_{k|k-1(0,0)}(P_{k|k-1(1,1)} + R_{(1,1)}) - P_{k|k-1(0,1)}P_{k|k-1(0,1)} & P_{k|k-1(0,1)}R_{(0,0)} \\ P_{k|k-1(0,1)}R_{(1,1)} & -P_{k|k-1(0,1)}P_{k|k-1(0,1)} + P_{k|k-1(1,1)}(P_{k|k-1(0,0)} + R_{(0,0)}) \\ P_{k|k-1(0,2)}(P_{k|k-1(1,1)} + R_{(1,1)}) - P_{k|k-1(1,2)}P_{k|k-1(0,1)} & -P_{k|k-1(0,1)}P_{k|k-1(0,2)} + P_{k|k-1(1,2)}(P_{k|k-1(0,0)} + R_{(0,0)}) \\ P_{k|k-1(0,3)}(P_{k|k-1(1,1)} + R_{(1,1)}) - P_{k|k-1(1,3)}P_{k|k-1(0,1)} & -P_{k|k-1(0,3)}P_{k|k-1(0,1)} + P_{k|k-1(1,3)}(P_{k|k-1(0,0)} + R_{(0,0)}) \end{bmatrix}$$

提取公因子（注意这里为命名一致，PR01 的命名跟随前面 PR00,PR11，有一个 R）：

$$PR_{00} = P_{k|k-1(0,0)} + R_{(0,0)}, PR_{11} = P_{k|k-1(1,1)} + R_{(1,1)}, PR_{01} = P_{k|k-1(0,1)}P_{k|k-1(0,1)}$$

式子可以改为

$$K_k = \frac{1}{PR_{00}PR_{11} - PR_{01}} * \begin{bmatrix} P_{k|k-1(0,0)}PR_{11} - PR_{01} & P_{k|k-1(0,1)}R_{(0,0)} \\ P_{k|k-1(0,1)}R_{(1,1)} & -PR_{01} + P_{k|k-1(1,1)}PR_{00} \\ P_{k|k-1(0,2)}PR_{11} - P_{k|k-1(1,2)}P_{k|k-1(0,1)} & -P_{k|k-1(0,1)}P_{k|k-1(0,2)} + P_{k|k-1(1,2)}PR_{00} \\ P_{k|k-1(0,3)}PR_{11} - P_{k|k-1(1,3)}P_{k|k-1(0,1)} & -P_{k|k-1(0,3)}P_{k|k-1(0,1)} + P_{k|k-1(1,3)}PR_{00} \end{bmatrix}$$

```
//计算Kk
const float PR00=P_k_k_prev_0_0+R_0_0;
const float PR11=P_k_k_prev_1_1+R_1_1;
const float PR01=P_k_k_prev_0_1*P_k_k_prev_0_1;
const float det=1.f/(PR00*PR11-PR01);
K_k[0][0]=det*(P_k_k_prev_0_0*PR11-PR01);
K_k[0][1]=det*(P_k_k_prev_0_1*R_0_0);
K_k[1][0]=det*P_k_k_prev_0_1*R_1_1;
K_k[1][1]=det*(-PR01+P_k_k_prev_1_1*PR00);
K_k[2][0]=det*(P_k_k_prev_0_2*PR11-P_k_k_prev_1_2*P_k_k_prev_0_1);
K_k[2][1]=det*(-P_k_k_prev_0_1*P_k_k_prev_0_2+P_k_k_prev_1_2*PR00);
K_k[3][0]=det*(P_k_k_prev_0_3*PR11-P_k_k_prev_1_3*P_k_k_prev_0_1);
K_k[3][1]=det*(-P_k_k_prev_0_3*P_k_k_prev_0_1+P_k_k_prev_1_3*PR00);
```

后验估计

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(z_k - h(\tilde{x}_{k-1}, 0)) = \hat{x}_{k|k-1} + K_k(y_k - C\tilde{x}_{k-1})$$

其中

$$\tilde{x}_{k-1} = f(\hat{x}_{k-1|k-1}, u_k, 0) = \hat{x}_{k|k-1}$$

```
//ypCx=y-Cx~(k-1)=y-Cf(xk-1|k-1,uk,0)=y-Cxk|k-1
float ypCx[2] = {
	ekf->Ialpha_I - x_k_k_prev[0],
	ekf->Ibeta_I - x_k_k_prev[1]
};
for (uint8_t a = 0; a < 4; ++a)
	x_k_k[a] = x_k_k_prev[a] + K_k[a][0] * ypCx[0] + K_k[a][1] * ypCx[1];
```

更新协方差矩阵

$$P_{k|k} = (I - K_kC)P_{k|k-1}$$

$$K_kCP_{k|k-1} = \begin{bmatrix} K_{k(0,0)} & K_{k(0,1)} & 0 & 0 \\ K_{k(1,0)} & K_{k(1,1)} & 0 & 0 \\ K_{k(2,0)} & K_{k(2,1)} & 0 & 0 \\ K_{k(3,0)} & K_{k(3,1)} & 0 & 0 \end{bmatrix} \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} & P_{k|k-1(0,2)} & P_{k|k-1(0,3)} \\ P_{k|k-1(0,1)} & P_{k|k-1(1,1)} & P_{k|k-1(1,2)} & P_{k|k-1(1,3)} \\ P_{k|k-1(0,2)} & P_{k|k-1(1,2)} & P_{k|k-1(2,2)} & P_{k|k-1(2,3)} \\ P_{k|k-1(0,3)} & P_{k|k-1(1,3)} & P_{k|k-1(2,3)} & P_{k|k-1(3,3)} \end{bmatrix} =$$

$$K_k C P_{k|k-1} = \begin{bmatrix} K_{k(0,0)} & K_{k(0,1)} \\ K_{k(1,0)} & K_{k(1,1)} \\ K_{k(2,0)} & K_{k(2,1)} \\ K_{k(3,0)} & K_{k(3,1)} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} & P_{k|k-1(0,2)} & P_{k|k-1(0,3)} \\ P_{k|k-1(0,1)} & P_{k|k-1(1,1)} & P_{k|k-1(1,2)} & P_{k|k-1(1,3)} \\ P_{k|k-1(0,2)} & P_{k|k-1(1,2)} & P_{k|k-1(2,2)} & P_{k|k-1(2,3)} \\ P_{k|k-1(0,3)} & P_{k|k-1(1,3)} & P_{k|k-1(2,3)} & P_{k|k-1(3,3)} \end{bmatrix}$$

$$= \begin{bmatrix} K_{k(0,0)} & K_{k(0,1)} \\ K_{k(1,0)} & K_{k(1,1)} \\ K_{k(2,0)} & K_{k(2,1)} \\ K_{k(3,0)} & K_{k(3,1)} \end{bmatrix} \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} & P_{k|k-1(0,2)} & P_{k|k-1(0,3)} \\ P_{k|k-1(0,1)} & P_{k|k-1(1,1)} & P_{k|k-1(1,2)} & P_{k|k-1(1,3)} \end{bmatrix}$$

这里无法再继续化简了

$$P_{k|k} = \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} & P_{k|k-1(0,2)} & P_{k|k-1(0,3)} \\ P_{k|k-1(0,1)} & P_{k|k-1(1,1)} & P_{k|k-1(1,2)} & P_{k|k-1(1,3)} \\ P_{k|k-1(0,2)} & P_{k|k-1(1,2)} & P_{k|k-1(2,2)} & P_{k|k-1(2,3)} \\ P_{k|k-1(0,3)} & P_{k|k-1(1,3)} & P_{k|k-1(2,3)} & P_{k|k-1(3,3)} \end{bmatrix} - \begin{bmatrix} K_{k(0,0)} & K_{k(0,1)} \\ K_{k(1,0)} & K_{k(1,1)} \\ K_{k(2,0)} & K_{k(2,1)} \\ K_{k(3,0)} & K_{k(3,1)} \end{bmatrix} \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} & P_{k|k-1(0,2)} & P_{k|k-1(0,3)} \\ P_{k|k-1(0,1)} & P_{k|k-1(1,1)} & P_{k|k-1(1,2)} & P_{k|k-1(1,3)} \end{bmatrix}$$

```
//更新协方差矩阵

P_k_k_0_0=P_k_k_prev_0_0-K_k[0][0]*P_k_k_prev_0_0-K_k[0][1]*P_k_k_prev_0_1;
P_k_k_0_1=P_k_k_prev_0_1-K_k[0][0]*P_k_k_prev_0_1-K_k[0][1]*P_k_k_prev_1_1;
P_k_k_0_2=P_k_k_prev_0_2-K_k[0][0]*P_k_k_prev_0_2-K_k[0][1]*P_k_k_prev_1_2;
P_k_k_0_3=P_k_k_prev_0_3-K_k[0][0]*P_k_k_prev_0_3-K_k[0][1]*P_k_k_prev_1_3;

P_k_k_1_1=P_k_k_prev_1_1-K_k[1][0]*P_k_k_prev_0_1-K_k[1][1]*P_k_k_prev_1_1;
P_k_k_1_2=P_k_k_prev_1_2-K_k[1][0]*P_k_k_prev_0_2-K_k[1][1]*P_k_k_prev_1_2;
P_k_k_1_3=P_k_k_prev_1_3-K_k[1][0]*P_k_k_prev_0_3-K_k[1][1]*P_k_k_prev_1_3;

P_k_k_2_2=P_k_k_prev_2_2-K_k[2][0]*P_k_k_prev_0_2-K_k[2][1]*P_k_k_prev_1_2;
P_k_k_2_3=P_k_k_prev_2_3-K_k[2][0]*P_k_k_prev_0_3-K_k[2][1]*P_k_k_prev_1_3;

P_k_k_3_3=P_k_k_prev_3_3-K_k[3][0]*P_k_k_prev_0_3-K_k[3][1]*P_k_k_prev_1_3;
```

经优化后，一次 EKF 大概需要 17us，可以满足 20khz 的需求